

Data Link Control in Computer Networks



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What is Data Link Control?

- Definition: Data Link Control (DLC) manages reliable node-to-node transfer across a single link.
- part of the Data Link Layer that manages communication between two directly connected devices. It helps data move in an organized, reliable, and error-free way.
- Purpose: Framing, synchronization, error detection/correction, and access control for link-level reliability.

Core Functions of Data Link Control



Line Discipline

Controls who transmits and when (channel access).



Flow Control

Manages sender/receiver speed to avoid overflow.



Error Control

Detects and corrects transmission errors.



Framing

Groups bits into frames with clear boundaries.

Line Discipline

Line discipline decides **which device can send data and when it can send it**

ENQ / ACK (Point-to-Point)

Sender sends ENQ (enquiry). Receiver replies with ACK to grant permission, then sender transmits frame(s).



Poll / Select (Multipoint)

Primary polls each secondary; primary selects which secondary may transmit. Prevents collisions on shared media.



Flow Control

Flow control prevents the sender from overwhelming the receiver's buffer and keeps link utilization efficient.

Flow control makes sure the sender does not send data faster than the receiver can handle.

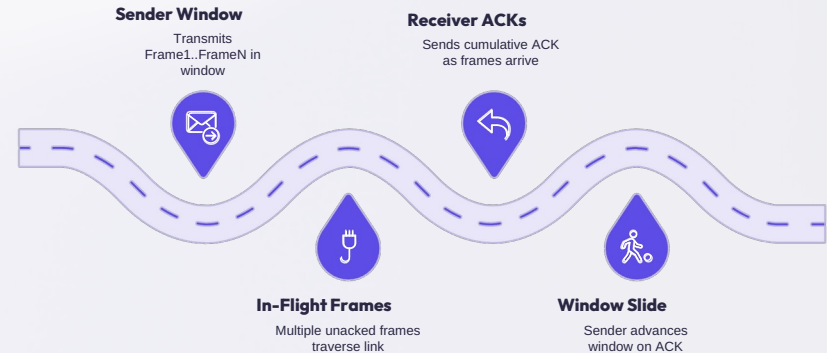
Stop-and-Wait

Sender transmits one frame then waits for ACK before sending the next. Simple, reliable, but limits throughput on high-latency links.



Sliding Window

Sender may transmit multiple frames up to window size without waiting for each ACK. Uses sequence numbers and cumulative ACKs for efficiency.



Error Control

Handles errors introduced by the physical layer: detects corrupted frames and (when possible) corrects them to preserve data integrity.

Types of Errors

- Single-bit error — isolated bit flips.
- Burst error — contiguous sequence of bit errors (often from noise).

Detection Methods

- **Parity Check** — adds an extra parity bit to detect odd/even bit errors.
- **CRC (Cyclic Redundancy Check)** — polynomial-based remainder; excellent detection for burst errors.

Correction: Hamming Code can locate and correct single-bit errors by adding redundant parity bits.

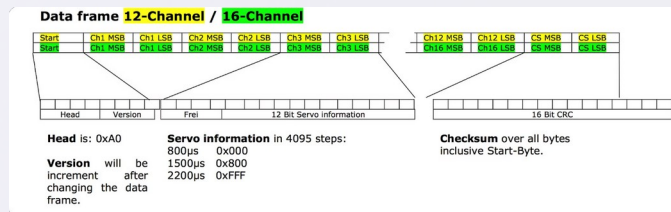
Detection

Parity (fast), CRC
(robust for bursts)

Correction

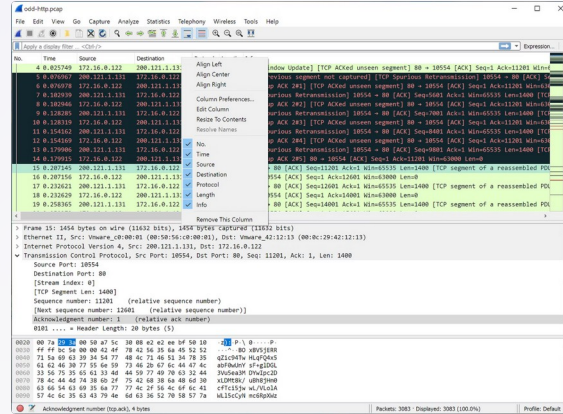
Hamming Code
(single-bit correction),
ARQ schemes
(retransmit on
detection)

Visual: small diagram shows data block, CRC generator dividing polynomial, remainder attached as checksum.



Framing

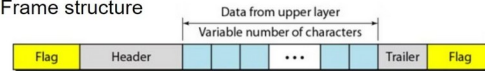
Framing divides the continuous bit stream into identifiable frames so receivers can parse and process data units correctly.



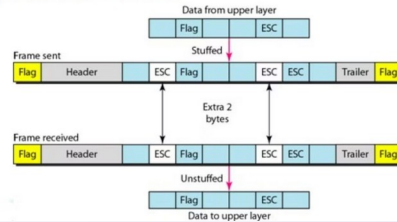
Character Count

Header field specifies the frame length. Simple but vulnerable to errors in the count field (loss/bit flips corrupt frame boundaries).

• Frame structure



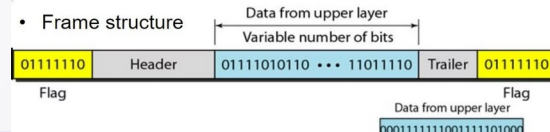
• Byte stuffing: process of adding 1 extra byte whenever there is a flag or escape character in the text



Flag Bytes + Byte Stuffing

Flag byte (e.g., 0x7E) marks frame start/end. If flag appears in data, an escape/extra byte is inserted.

• Frame structure



Conclusion

- Line Discipline: Controls channel access (ENQ/ACK for point-to-point, Poll/Select for multipoint).
- Flow Control: Prevents overflow via Stop-and-Wait or Sliding Window.
- Error Control: Detects errors (Parity, CRC) and corrects them (Hamming Code, ARQ).
- Framing: Turns raw bits into frames using count fields, byte/bit stuffing, or coding violations.

Together, these four functions make data link communication reliable and efficient.

Thank You for Listening!

Any Questions?

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